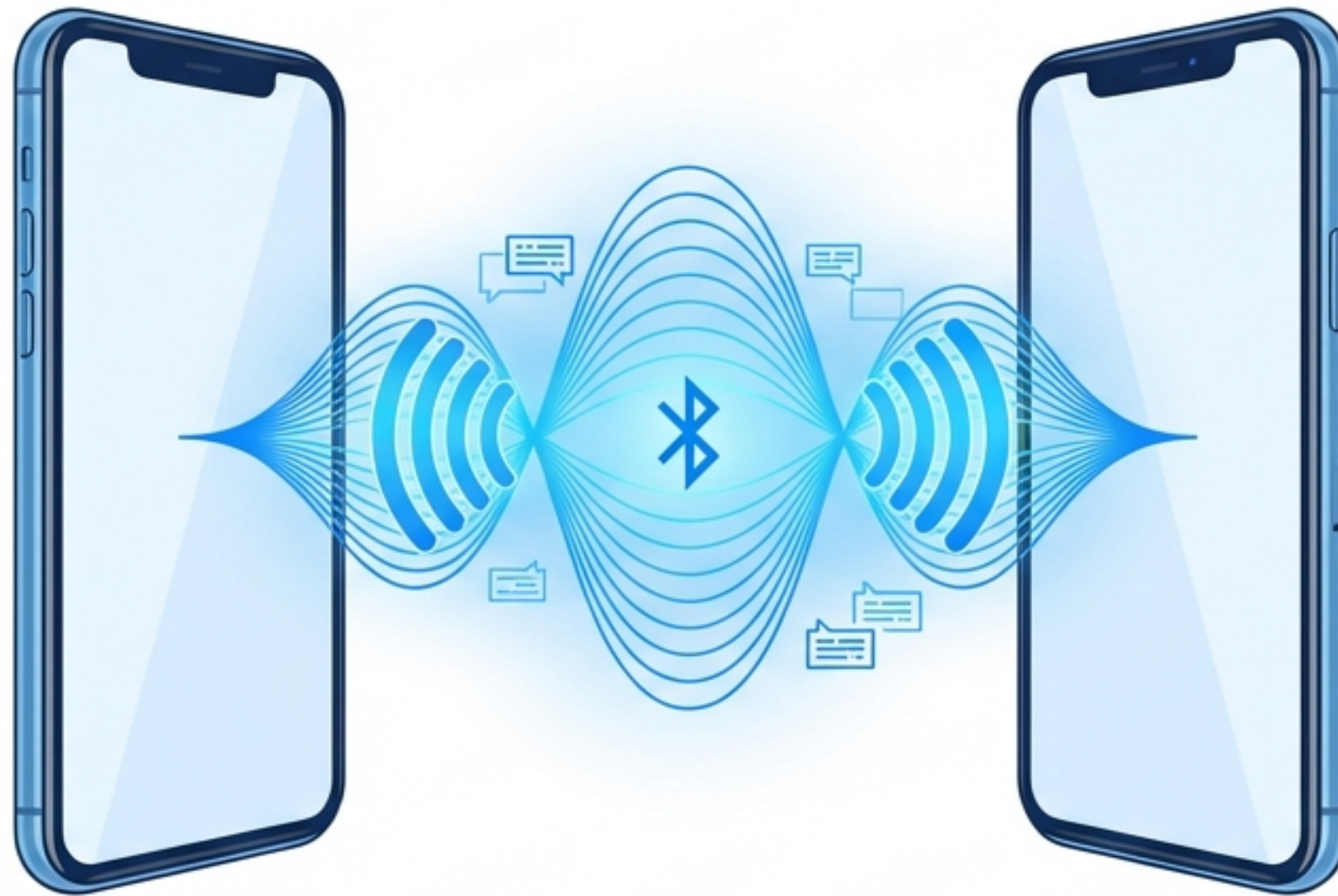


Bluetooth Communication

ICT 360 – Grade 7 | Connecting Without the Internet
Theory • Tools • Action

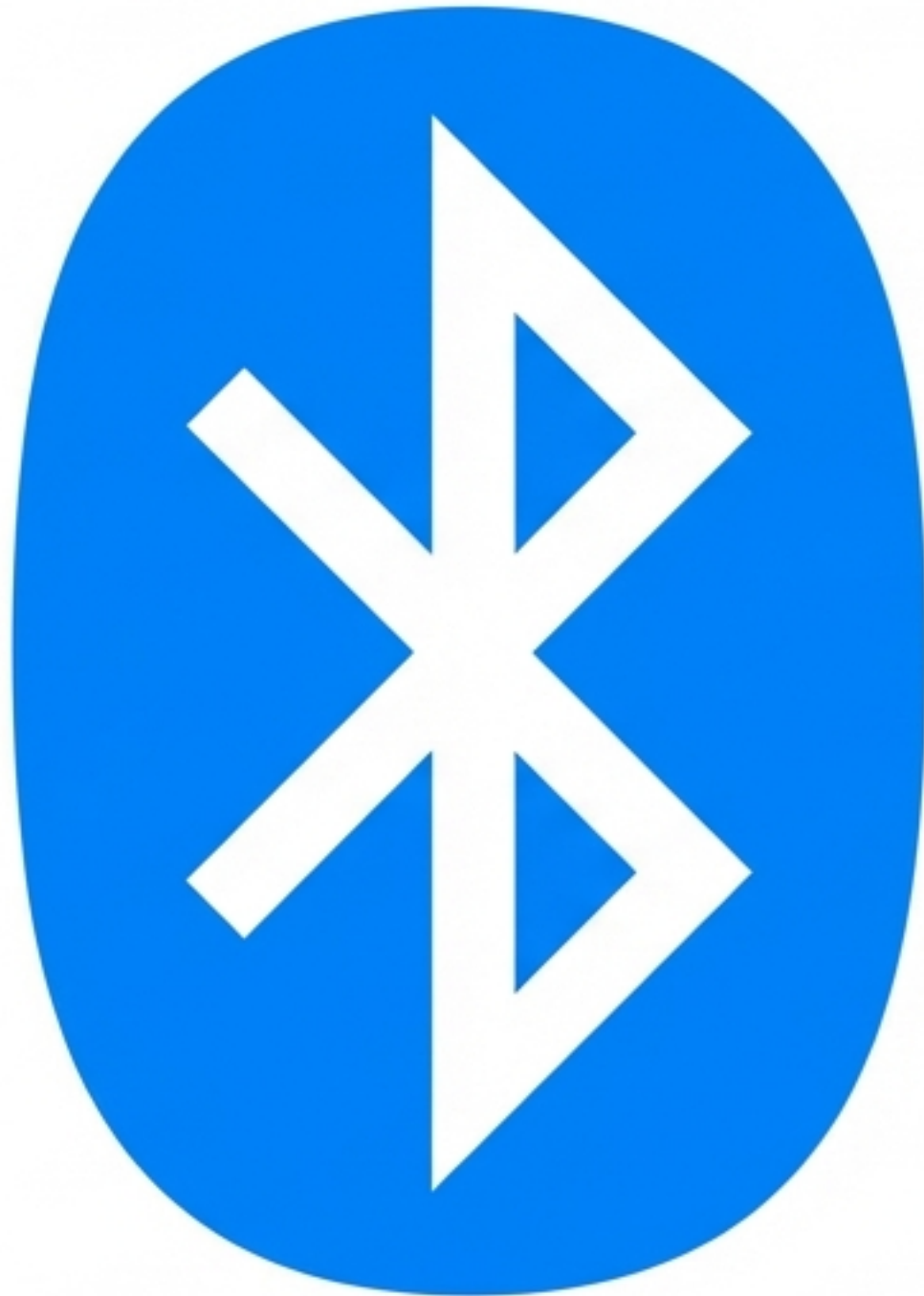


The World of Apps

- **What is an App?** Software that performs specific tasks (games, maps, chat).
- **The Problem:** Most apps rely on the Internet to work.
- **The Question:** How do we share data when the Wi-Fi is down?
- **The Answer:** Short-distance wireless sensors (Bluetooth).

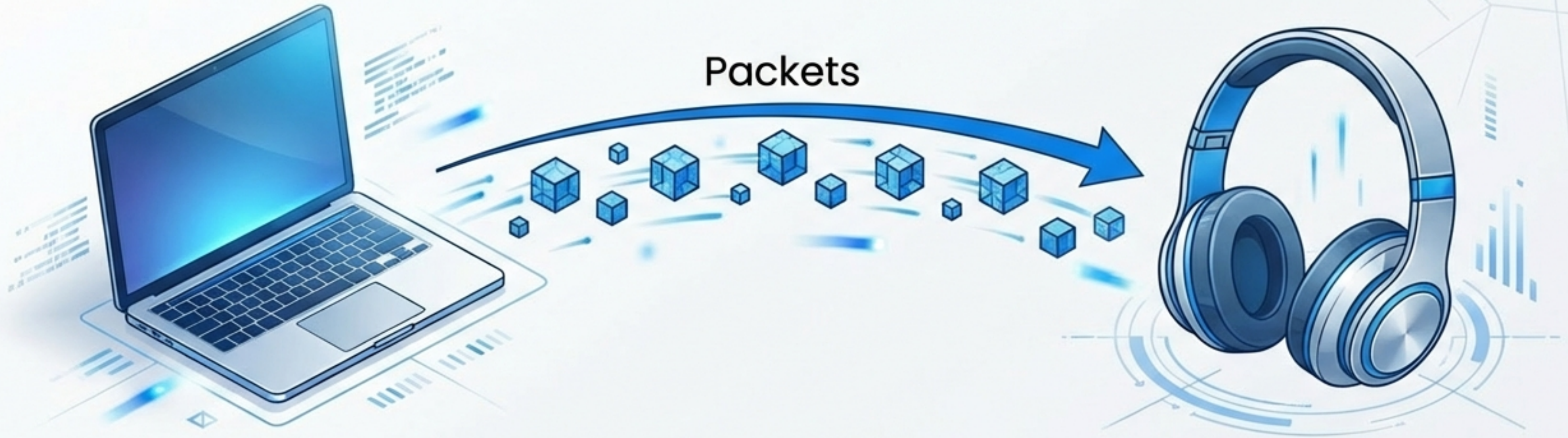


Meet Bluetooth



- **Definition:** Open wireless technology for exchanging data over short distances.
- **Network:** Creates a secure Personal Area Network (PAN).
- **Origin Story:** Named after King Harald 'Bluetooth' Gormsson, who united Scandinavia. Just like the King, this technology unites our devices!

The Science of Connection



Frequency

Operates on radio waves at 2.4 GHz.

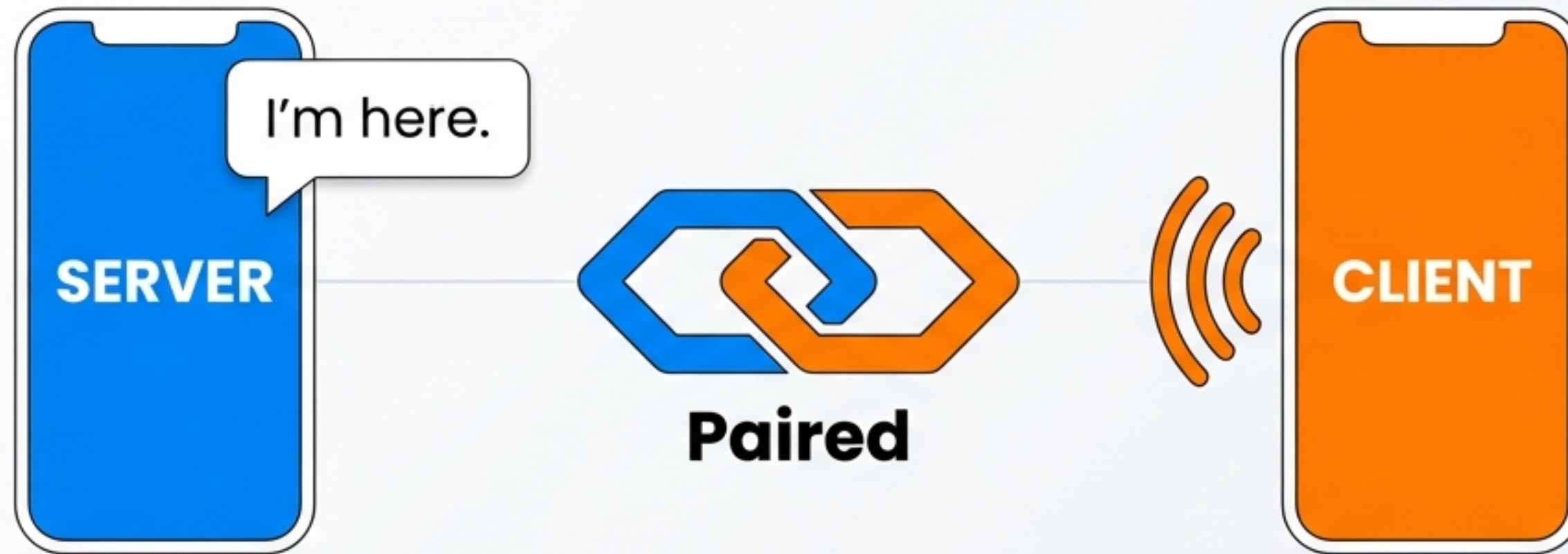
Packets

Data is sliced into small pieces called 'Packets' to be sent wirelessly.

Channels

Devices hop between 79 different channels to avoid traffic jams.

The Connection Dance: Server & Client



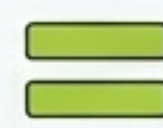
- **Server:** The Listener. It waits for a connection and accepts it.
 - **Client:** The Searcher. It looks for the server and requests to connect.
- 📝 **Note:** In our project, we will build BOTH apps!



Designer Window
(The Look)



Blocks Editor
(The Logic)



Real Android App

Our Toolkit: MIT App Inventor

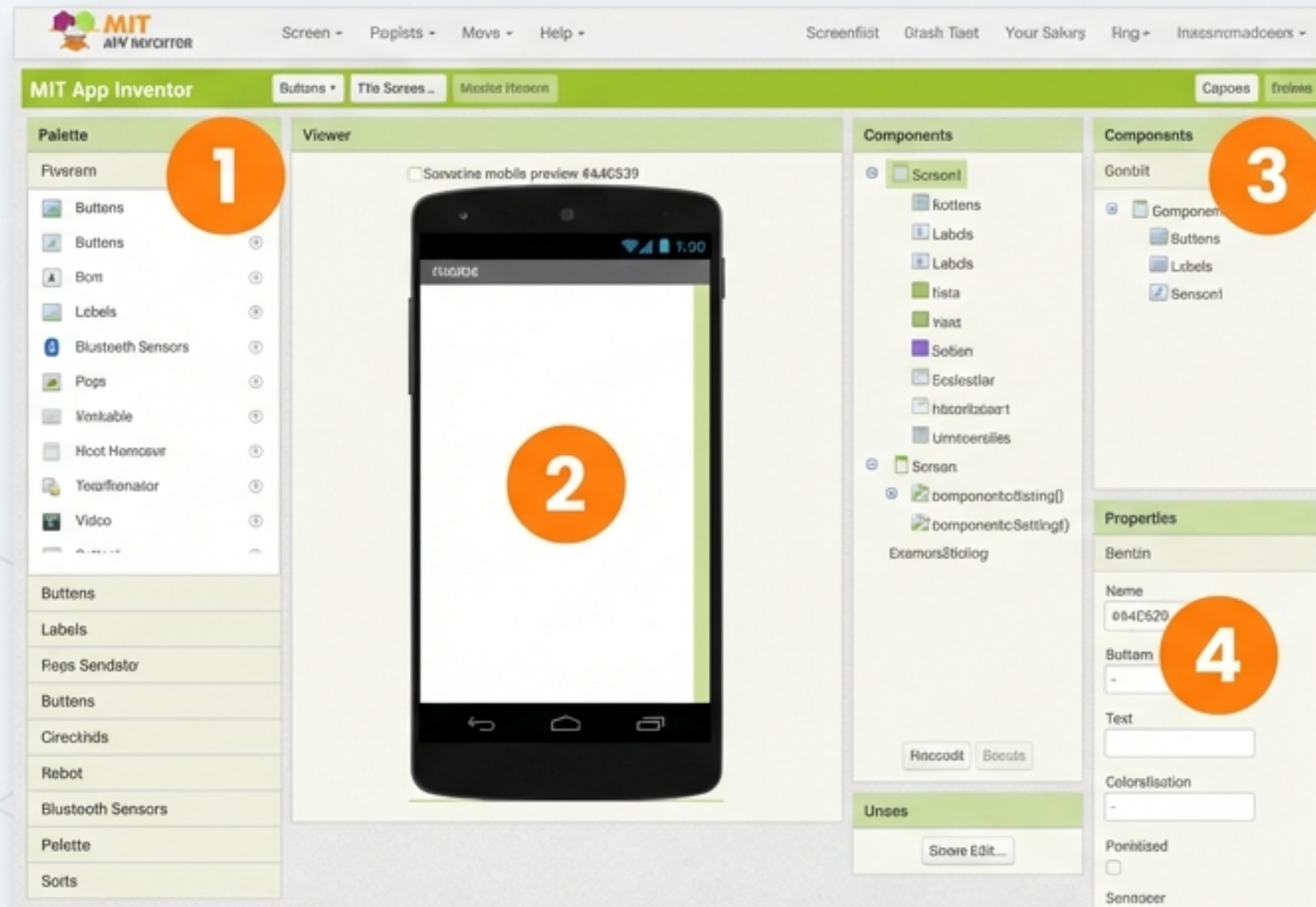
What is it? A visual tool to build real Android apps. 

Why use it?* No complex typing—just drag and drop logic blocks! 

Two Main Views:

- 1. **Designer:** Where you draw how the app looks. 
- 2. **Blocks Editor:** Where you decide what the app does. 

Knowing Your Workspace



- 1. Palette:** The toolbox containing Buttons, Labels, and Bluetooth Sensors.
- 2. Viewer:** Your canvas. This shows what the app will look like.
- 3. Components:** A list of everything you have added to the screen.
- 4. Properties:** Settings to change colors, font sizes, and text.

The Mission: Build a Bluetooth Chat

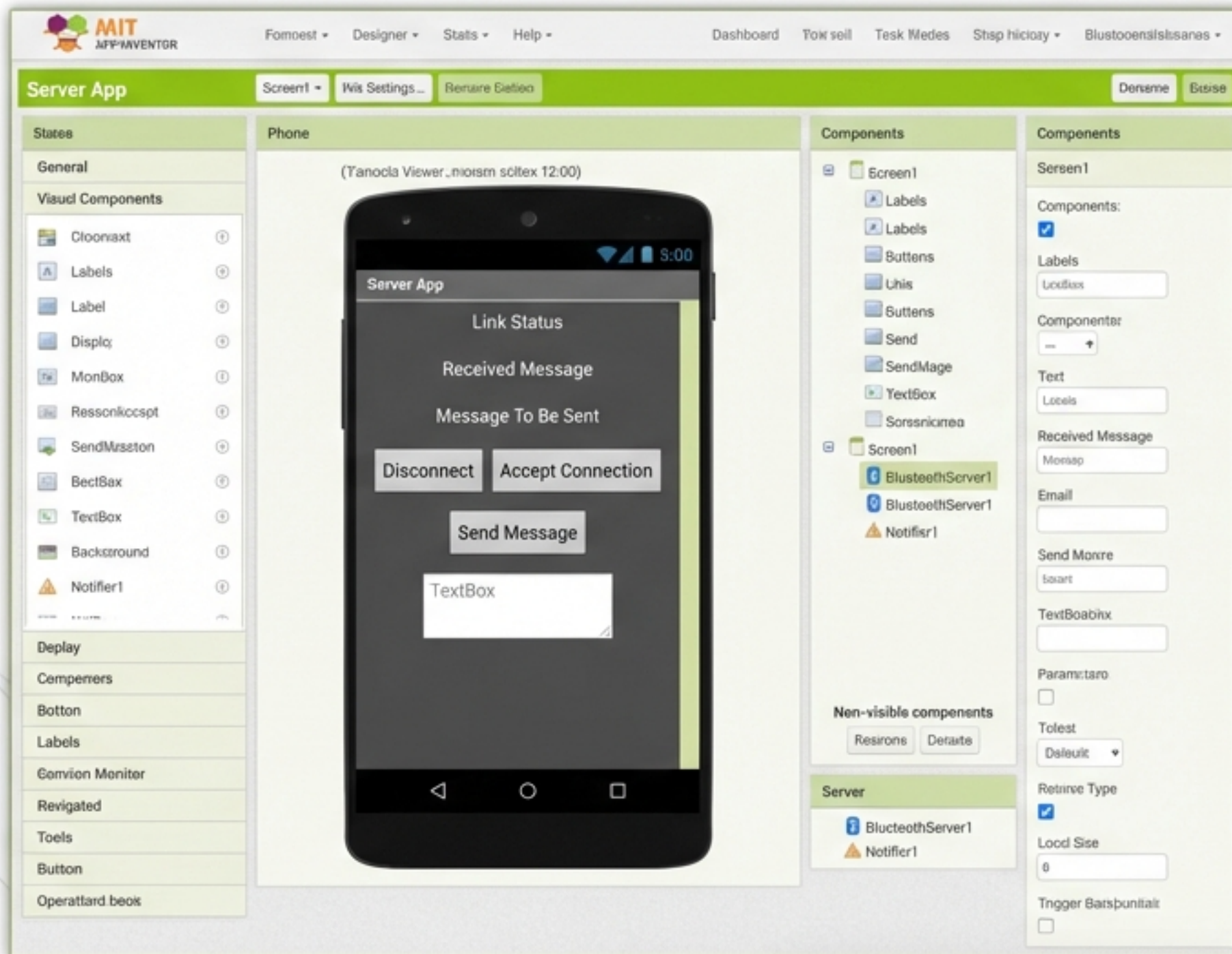


The Goal: Create a 2-way communication system between two smartphones.

The Plan:

- 1. Build the **Server App**.
- 2. Build the **Client App**.
- 3. Connect and Chat!

Step 1: Designing the Server (UI)



- **Visual Components:**
 - Labels: 'Link Status', 'Received Message'
 - Buttons: 'Accept Connection', 'Disconnect', 'Send'
 - TextBox: To type your message
- **Invisible Components (Crucial!):**
 - BluetoothServer1: To handle the connection.
 - Notifier1: To show pop-up alerts.

Design Tip: Use Dark Gray backgrounds and White text for a professional look.

Step 2d 2: Coding the Server (Logic)

The Listener:

When clicked, it opens the door for a client.

```
when Screen1.Initialize
do
  if not BluetoothServer1.Enabled
  then
    call Notifier1.ShowAlert
      notice "Enable Bluetooth first"
```

```
when Accept_Connection.Click
do
  call BluetoothServer1.AcceptConnection
    serviceName ""
  set Status.Text to "Accepting Connection"
```

```
when BluetoothServer1.ConnectionAccepted
do
  set Status.Text to "Connection Accepted"
```

```
when BluetoothServer1.BluetoothError
  functionName message
do
  set Status.Text to get message
```

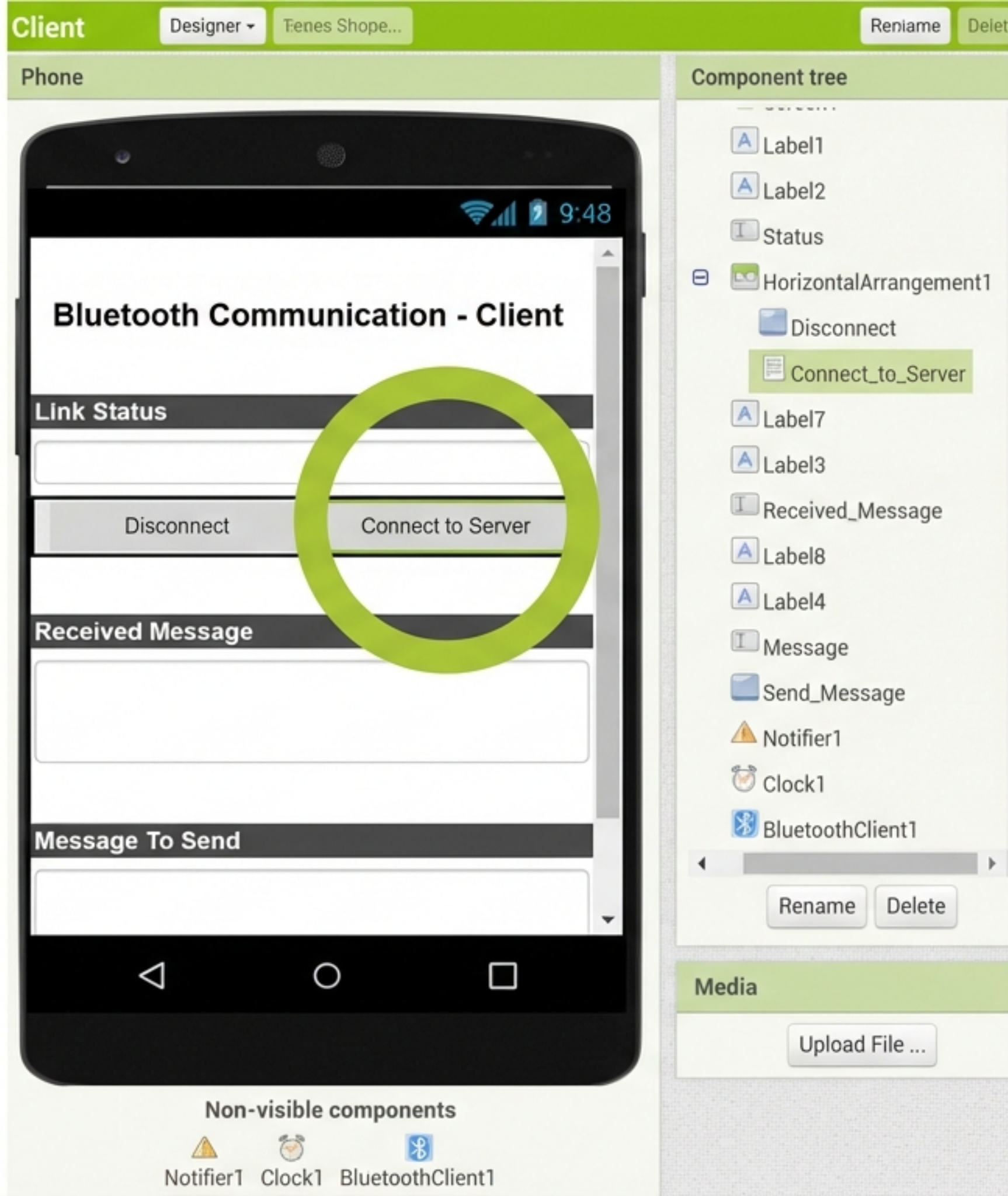
The Timer:

Constantly checks: Is there new data coming in?

```
when Clock1.Timer
do
  if BluetoothServer1.IsConnected
  then
    if call BluetoothServer1.BytesAvailableToReceive > 0
    then
      set Received_Message.Text to call BluetoothServer1.ReceiveText
        numberOfBytes call BluetoothServer1.BytesAvailableToReceive
```

The Display:

If data arrives, show it on the screen.



Step 3: Designing the Client (UI)

- **The Key Difference:** The Server waits, but the Client must **choose** who to talk to.
- **The ListPicker:** Instead of a normal button, we use a **ListPicker** component for 'Connect to Server'. This opens a menu of available Bluetooth devices.
- **Invisible Component:** Add **BluetoothClient1** (Note: Client, not Server!).

Step 4: Coding the Client (Logic)

Before Picking:
Show a list of all
paired devices
(Addresses and
Names).

```
when Screen1.Initialize
do set BluetoothClient to "bant"

when Connect_to_Server.BeforePicking
do set Connect_to_Server.Devices to (Addresses to Names)

when Connect_to_Server.AfterPicking
do call Connect_to_Server.Pickicking user get Connect_to_Server

when Screen1.ErrorOccurred
miudal natos erroroccured
do

when Send_Message.Click
do set Send_Message.When the text get Send_Message

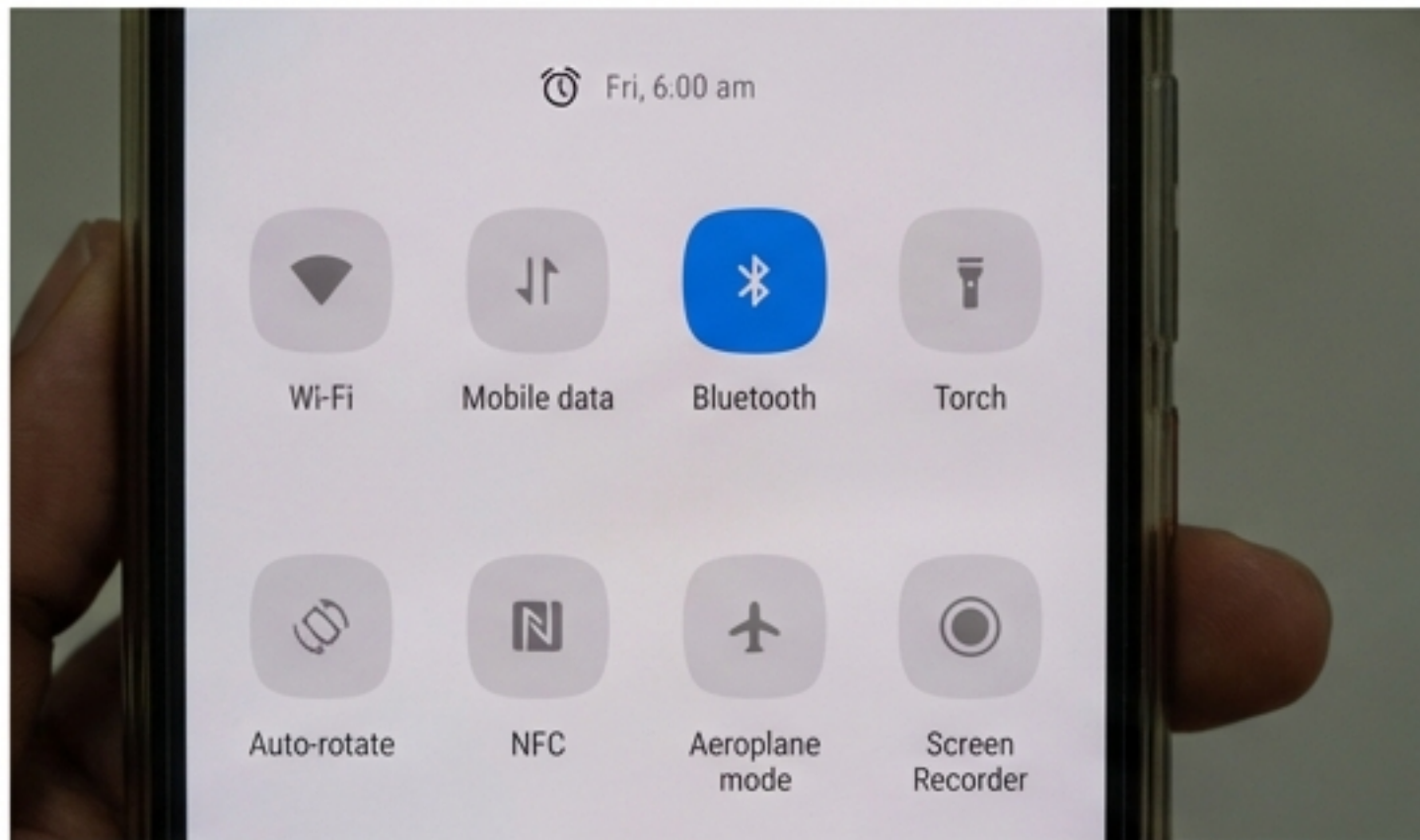
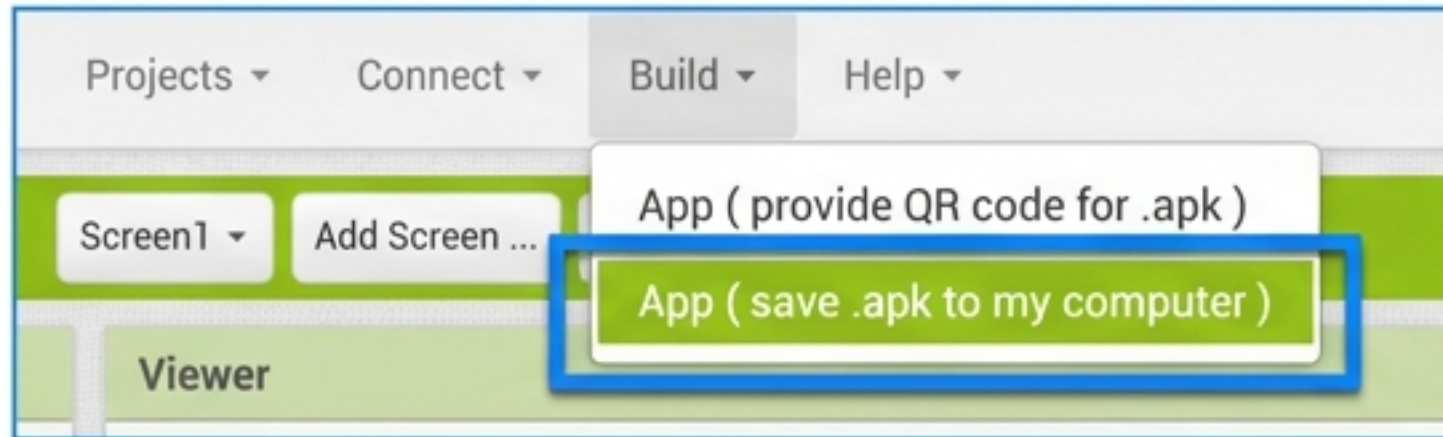
when Disconnect.Click
do set DisconnectClient to "Disconnect"

when Clock1.Timer
do set Clock1.Text to lms
```

After Picking:
Connect to the
specific address
chosen by the user.

Sending Data: When
"Send" is clicked, push
the text through the
Bluetooth pipe.

Bringing It to Life



1. **Build:** Save the .apk file to your computer.
2. **Install:** Transfer the file to two different phones via USB.
3. **Setup:** Turn on Bluetooth on BOTH phones.
4. **Run:** Open Server on Phone A, Client on Phone B. Pair and Chat!

Tech Flash: What We Learned



PAN

(Personal Area Network)



Packets

(Chunks of data sent wirelessly)



Server / Client

(The Listener and the Initiator)



MIT App Inventor

(The block coding tool)

You Are Now a Developer!



Mission Accomplished: You turned a phone into a custom communication tool.

Next Challenge: Bluetooth connects nearby devices. What else could you control?

- A robot?
- A multiplayer game?
- A home automation system?

Thank You!